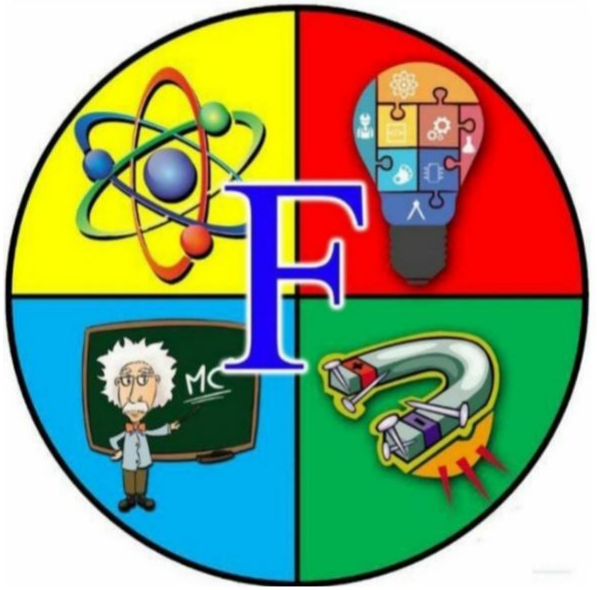
Fizika

9-sinf

Mavzu: **Masalalar yechish**



29-mashq 1-masala (157-bet)

Balandligi 3 m bo‘lgan daraxt suratga olinganda uning tasvirining balandligi 12 mm bo‘ldi. Agar fotoapparat obyektivining fokus masofasi 20 cm bo‘lsa, surat qanday masofadan olingan?

Berilgan:

$H = 3\ m$

$h = 12\ mm = 0{,}012\ m$

$F = 20\ cm = 0{,}2\ m$

Topish kerak: $d-?$

Formula:

$$\frac{1}{F} = \frac{1}{d} + \frac{1}{f}$$

$$\frac{1}{F} = \frac{f+d}{df}$$

$$K = \frac{h}{H} = \frac{f}{d}$$

$$f = \frac{hd}{H}$$

$$\frac{1}{F} = \frac{\frac{hd}{H}+d}{d\frac{hd}{H}} \Rightarrow \frac{1}{F} = \frac{\frac{hd+Hd}{H}}{\frac{hd^2}{H}} \Rightarrow \frac{1}{F} = \frac{(h+H)d}{hd^2} \Rightarrow \frac{1}{F} = \frac{h+H}{hd}$$

$$d = \frac{F(h+H)}{h}$$

$$d = \frac{F(h+H)}{h}$$

Yechish: $d = \frac{0{,}2\ m \cdot (3\ m + 0{,}012\ m)}{0{,}012\ m} = 50{,}2\ m$

Javob: $d = 50{,}2\ m$

29-mashq 2-masala (157-bet)

Mikroskop ob’yektivining fokus masofasi 2 mm, okulyarining fokus masofasi 30 mm. Okulyar bilan fokus orasidagi masofa 20 cm bo‘lsa, mikroskopning kattalashtirishini toping.

Berilgan:

$F_1 = 2\ mm = 0{,}002\ m$

$F_2 = 30\ mm = 0{,}03\ m$

$l = 20\ cm = 0{,}2\ m$

$D_0 = 25\ cm = 0{,}25\ m$

Topish kerak: $K - ?$

Formula:

$$K = \frac{lD_0}{F_1 F_2}$$

Yechish:

$$K = \frac{0{,}2\ m \cdot 0{,}25\ m}{0{,}002\ m \cdot 0{,}03\ m} = 833{,}3$$

Javob: $K = 833,3$

29-mashq 5-masala (157-bet)

O‘quvchi bola ko‘zoynagini olib, kitobni ko‘zidan 16 cm masofada o‘qiydi. U taqqan ko‘zoynakning optik kuchi qanday bo‘lgan?

Berilgan:

$d = 16\ cm = 0{,}16\ m$

$d_0 = 25\ cm = 0{,}25\ m$

Topish kerak: $D - ?$

Formula:

$$D = \frac{1}{d_0} - \frac{1}{d}$$

Yechish: $D = \frac{1}{0{,}25\ m} - \frac{1}{0{,}16\ m} = -2{,}25\ dptr$

Javob: $\boldsymbol{D = -2{,}25\ dptr}$

29-mashq 6-masala (157-bet)

O‘quvchi optik kuchi $-2\ dptr$ bo‘lgan ko‘zoynakda o‘qimoqda. Uning ko‘zoynaksiz eng yaxshi ko‘rish masofasi qanday?

Berilgan:

$D = -2\ dptr$

$d_0 = 25\ cm = 0{,}25\ m$

Topish kerak: $d - ?$

Formula:

$$D = \frac{1}{d_0} - \frac{1}{d}$$

$$\frac{1}{d} = \frac{1}{d_0} - D$$

$$\frac{1}{d} = \frac{1 - D d_0}{d_0}$$

$$d = \frac{d_0}{1 - D d_0}$$

Yechish: $d = \frac{0{,}025\ m}{1-(-2 dptr)\cdot 0{,}25} \approx 0{,}167\ m \approx 16{,}7\ cm$

Javob: $\boldsymbol{d \approx 16{,}7\ cm}$

29-mashq 7-masala (157-bet)

Havoga nisbatan nur sindirish koʻrsatkichi 1,5 ga teng shishadan yasalgan sirtining egrilik radiuslari 25 cm va 40 cm boʻlgan ikki yoqlama qavariq linzaning optik kuchini toping.

Berilgan:

$n = 1{,}5$

$R_1 = 25\ cm = 0{,}25\ m$

$R_2 = 40\ cm = 0{,}4\ m$

Topish kerak: $D - ?$

Formula:

$$F = \frac{1}{(n-1)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)};\ F = \frac{1}{D}$$

$$\frac{1}{D} = \frac{1}{(n-1)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)}$$

$$D = (n-1)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$$

Yechish: $D = (1{,}5-1)\cdot\left(\frac{1}{0{,}25\ m}+\frac{1}{0{,}4\ m}\right) = 3{,}25\ dptr$

Javob: $\boldsymbol{D = 3{,}25\ dptr}$

Mustaqil bajarish uchun topshiriqlar

1. Havoga nisbatan nur sindirish koʻrsatkichi 2 ga teng moddadan yasalgan sirtining egrilik radiuslari 20 cm va 25 cm boʻlgan ikki yoqlama qavariq linzaning optik kuchini toping.

2. Oʻquvchi bola koʻzoynagini olib, kitobni koʻzidan 20 cm masofada oʻqiydi. U taqqan koʻzoynakning optik kuchi qanday boʻlgan?

3. 29-mashq 3-masalalarni yechish. (157-bet)